

DATA CRISIS — ROUND 2

THEME: “THE MODEL IS LYING” | COLLEGE TECHNICAL SYMPOSIUM

1. Executive Mission Brief & Scenario

You are deployed as an emergency Data Science Taskforce for **Apex Telecom**. Over recent quarters, the company has experienced elevated subscriber turnover. Your mission is to build a machine-learning system to predict customer churn (**churn = 1 vs 0**). However, **THE MODEL IS LYING**. The telemetry dataset was collected across disparate operational databases and contains severe data-quality problems, formatting anomalies, and hidden statistical traps. A naive model trained blindly will fool you during evaluation and fail on unseen test customers. You must perform data forensics, clean corrupted inputs, adapt to timed crisis broadcasts, and deliver a reliable production pipeline.

2. Provided Participant Files

File Name	Description
train.csv	Historical customer records (~2,400 rows) with target churn. Contains deliberate real-world data quality anomalies.
test.csv	Unseen test customer records (600 rows) without churn. You must generate predictions for these records.
submission_template.csv	Template showing the exact required schema: customer_id,churn (600 rows).
README.pdf / README.md	Official competition guidelines, column dictionary, and operational rules.

3. Event Flow & Timed Crisis Intervals (60–70 Mins Total)

Phase	Duration	Focus / Objectives
Phase 1	10 mins	Data Normalization: Investigate train.csv, detect missing values, duplicates, outliers, corrupt types, and inconsistent categories. Build cleaning scripts.
Phase 2	10 mins	Crisis Update 1: Riddle 1 broadcast. Decode the hidden flaw, investigate feature matrices, adapt pipeline.
Phase 3	10 mins	Crisis Update 2: Riddle 2 broadcast. Address class distribution challenges and objective metric traps.
Phase 4	25–30 mins	Crisis Update 3 & Final Modeling: Riddle 3 broadcast. Handle domain shifts, finalize model, export submission.csv.

4. Feature & Column Dictionary

Column Name	Type	Description / Domain Constraints
customer_id	String	Primary Key unique customer identifier (e.g. C0001, T0001).
age	Numeric	Subscriber age in years (expected range: 18 to 75).
gender	Categorical	Gender demographic (Male, Female).
tenure_months	Numeric	Consecutive subscription duration in months (1 to 72).
monthly_bill	Numeric/Text	Monthly telecom service charge (typical range: █299 – █1,999).
total_bill	Numeric/Text	Cumulative billing across account lifetime.
contract_type	Categorical	Service agreement type (Monthly, Annual, Two Year).
payment_method	Categorical	Payment channel (UPI, Credit Card, NetBanking, Debit Card).
internet_plan	Categorical	Broadband tier (Fiber, DSL, None).
support_calls	Integer	Customer service inquiries logged in the past 12 months.
complaints	Integer	Escalated formal tickets registered by subscriber.
usage_hours	Numeric	Active monthly service usage hours.
last_login_days	Integer	Days elapsed since last self-care portal access.
data_usage_gb	Numeric	High-speed data consumption in Gigabytes.

num_devices	Integer	Registered connected terminal devices (1 to 6).
discount_percent	Numeric/Text	Promotional invoice discount applied (0% to 25%).
late_payments	Integer	Count of overdue payments over customer tenure.
satisfaction_score	Integer	Customer satisfaction feedback rating (1=Worst, 10=Best).
region	Categorical	Operating zone (North, South, East, West).
plan_upgrade_count	Integer	Frequency of plan upgrades initiated by user.
paperless_billing	Categorical	Digital invoicing flag (Yes, No).
favorite_theme	Categorical	Mobile app interface preference (Dark, Light, System).
retention_offer_status	Categorical	Recorded status of retention intervention.
churn	Binary (0/1)	TARGET COLUMN (train.csv only): 1 = Churned, 0 = Stayed.

5. Qualification Gate, Scoring & AI Policy

PRIMARY REQUIREMENT: Accuracy > 90.00%

To qualify for the final round, your submission on the unseen test set must exceed 90.00% accuracy. Submissions with accuracy $\leq$ 90.00% are marked NOT QUALIFIED. Top 3 teams out of 6 advance.

SECONDARY RANKING FORMULA (For Qualifying Teams):

Final Score = **60% Accuracy + 20% F1-Score + 10% Recall + 10% Robustness Score**

AI USAGE POLICY: AI tools (ChatGPT, Gemini, Claude, Copilot, Documentation) are **PERMITTED**. However, teams must be prepared to defend and explain their preprocessing, feature selection, and model architecture. Prohibited: Accessing organizer files, hardcoding predictions, using external labeled datasets, or sharing code between teams.

6. Required Submission Schema

Deliverable: Single file named **submission.csv** with exactly 2 columns and 600 customer rows:

customer_id, churn

T0001,0

T0002,1

T0003,0

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